Forecast-Aware Trading and Control of Solar Generation with Battery Storage

Research framework, initial evidence and current directions

4 November 2026

Abstract

A solar producer must sell electricity before production and prices are fully known, then deliver an evolving contractual position with finite battery flexibility. We study how forecasts, sequential auction decisions and real-time physical control should work together. The framework revises future decisions while preserving past commitments and carrying actual stored energy and battery wear through time. Experiments cover four northern-Italian solar plants, day-ahead and intraday auctions, probabilistic solar forecasts, price prediction and several battery controllers. Early results show a recurring trade-off: reducing forecast error or imbalance does not necessarily maximize operating profit. The current focus is on useful uncertainty, adaptive control and the value of information under limited physical flexibility. Results remain retrospective, with short control studies and explicitly identified reconstructed production. Detailed equations are collected in the appendices.

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1 The problem: decisions made before uncertainty is resolved

A photovoltaic plant and a battery share a grid connection. Solar generation can be exported, stored or curtailed. The battery can also import energy for charging. There is no local demand. Its finite power, energy and conversion efficiency limit how much forecast error it can absorb, while cycling creates an economic wear cost.

The producer participates in the Italian day-ahead auction and three intraday auctions. Each new trade changes the quantity promised for a delivery period. The quantity becomes binding when submitted, even if its price has not yet been published. Later forecasts may change the preferred position, but earlier orders cannot be rewritten. At delivery, any difference between physical export and the contracted position is settled at the applicable imbalance price.

This creates two linked decisions: how much energy to contract at each auction, and how to operate the battery and solar plant as delivery approaches. Selling conservatively may reduce shortfall but leave valuable solar energy unused. Aggressive sales may earn more auction revenue but consume battery flexibility or create settlement exposure. Retaining energy today can protect tomorrow's operation, yet may forgo a profitable opportunity now.

We measure operating profit as auction receipts plus signed imbalance settlement, less executed battery wear, transaction fees and any modeled import charge. A positive imbalance is not automatically a loss: its cash sign depends on the settlement price. This is an operating model, not an investment valuation; capital cost, financing, taxes and fixed operating costs are outside its scope.

Auction	Gate, Italian local time	Delivery covered	Information update
Day ahead	Previous day, 12:00	Whole next day	Initial position
Intraday 1	Previous day, 15:00	Whole next day	Revise remaining trades
Intraday 2	Previous day, 22:00	Whole next day	Revise remaining trades
Intraday 3	Delivery day, 10:00	Noon to midnight	Revise afternoon trades

These are the study calendar's auction gates; decisions use a five-minute buffer. Quarter-hour delivery is represented with actual elapsed durations, including daylight-saving transitions. Earlier hourly day-ahead products remain single market products even when physical control uses quarter-hours. The market assumption is price-taking and full execution of submitted quantities. Continuous order-book trading and uncertain order acceptance are separate future problems.

2 Our core idea: connect forecasts, commitments and physical state

The framework repeatedly updates three connected layers.

1. **Forecast the available energy and its economic environment.** Use only information eligible at the decision time, with explicit assumptions about forecast release and observation delays. Represent solar uncertainty as possible paths rather than treating one point forecast as certain.
2. **Optimize the remaining auction opportunities.** Choose current and provisional future trades jointly with a physically feasible battery schedule. Preserve all submitted quantities. Execute only the order for the auction that is closing.
3. **Control delivery and carry the consequences forward.** Use observed battery state

and updated forecasts to choose physical actions. Execute only the first action, settle actual deviations, and use the resulting state in later decisions.

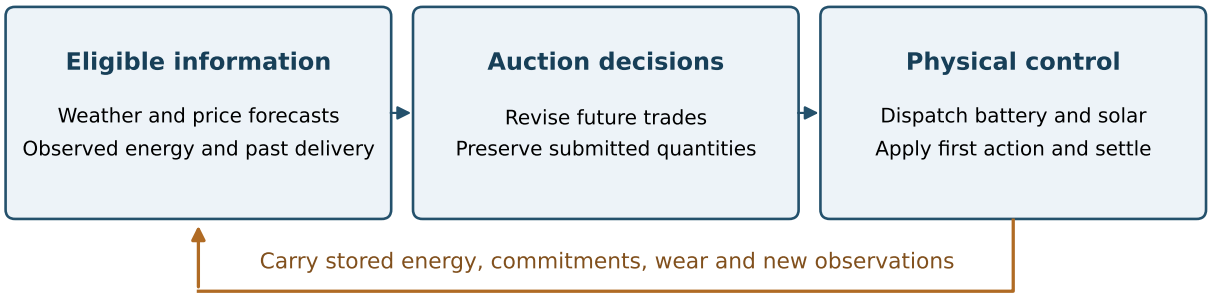


Figure 1: The auction layer revises future commitments; the physical layer operates against quantities already submitted. Actual state and observations connect successive decisions.

The key link is the *actual* battery state. A plan for tomorrow may be formed while today's delivery is still in progress. We project the boundary state using eligible forecasts and retained controls, but never overwrite the actual state with that prediction. When tomorrow arrives, its true carried state can differ. This keeps projection error, replanning and physical protection visible in the outcome.

Similarly, an optimization objective is an expectation over the remaining opportunities, not a cash account. Previously submitted trades remain binding; their revenue is booked once when prices are published. Repeated objectives over overlapping horizons are never added to estimate profit. In this work, continuous operation means this uninterrupted auction-and-control chronology, with daily planning horizons.

The research contribution is the integrated decision framework and its controlled comparisons: we study when better information changes economically useful decisions, how physical flexibility changes that value, and how the representation of uncertainty affects the action that can be taken now.

3 The main model and its extensions

The main auction model maximizes expected remaining trading revenue after wear and trading costs, subject to battery energy conservation, exclusive charging and discharging, grid limits and immutable earlier commitments. Decisions that have the same observed history must be identical across scenarios. The complete mixed-integer problem is in Appendix A.

The initial strict-delivery formulation requires each represented solar path to support the same contracted delivery and common battery schedule. Solar curtailment can adapt to available production. Under equal price paths, this has an exact simplification: useful solar injection cannot exceed the minimum across represented paths at each interval. It is a conservative planning constraint, not a new forecast trajectory or a calibrated confidence level.

A stochastic alternative allows deviations and penalizes expected absolute imbalance, optionally with a tail-volume term. Physical control then compares simple commitment tracking with receding-horizon control. Price-blind control balances deviation volume, wear and terminal energy. Forecast-economic control also values signed settlement, while a noncausal price oracle provides a separately labelled information benchmark.

The main extensions address distinct questions:

Adaptive future control.
All scenarios share the action taken now; future actions can branch only after distinguishable solar observations. This tests the value of adapting to information, rather than giving each path an unrealistically informed first action.

Coherent uncertainty through time.
Calibrated marginal distributions describe uncertainty at one time. Storage also responds to the duration of shortfalls and surpluses. Rank-based coupling preserves a raw ensemble's temporal structure, while matched shuffling tests what changes when that structure is removed.

Risk and remaining energy.
Monetary tail risk accounts for unfavorable operating-loss scenarios. A forecast-conditioned continuation value estimates the usefulness of energy remaining beyond the immediate control horizon. Neither term is actual income.

Information, market access and physical flexibility.
Factorial comparisons separate updated information from access to more auctions, and vary battery energy, battery power and connection capacity independently. This asks where forecast improvements can actually be used.

Smaller scenario sets.
Representative paths can reduce computation while preserving selected ramp, energy-deficit and economic features. Their quality must be assessed through decisions and outcomes; a small forecast-space distance is not an economic guarantee.

4 What has been tested

The four studied solar capacities are approximately 0.95, 1.52, 4.16 and 6.64 MW. The hypothetical batteries range from 0.75 to 5 MWh and 0.5 to 2.5 MW. They share a 7–93% state-of-charge range and 90% round-trip efficiency. Battery and connection ratings are study assumptions, not verified equipment inventories. The assumed throughput wear cost is approximately EUR 27.64 per AC MWh charged or discharged.

Forecast experiments include physical conversion/calibration, point neural models, marginal ensemble calibration, delayed online weighting, price-history and fundamental predictors, and native-resolution intraday spread models. Trading experiments compare auction participation, deterministic and ensemble planning, and several physical controllers. Each controller has its own chronological state and paired tracking baseline.

The evidence has three important boundaries. First, many auction studies use perfect future clearing prices to isolate physical and information effects; their profits are not implementable price-forecast expectations. Second, model selection reused parts of the historical evaluation period. Third, some production targets are reviewed reconstructions rather than independent measurements. We preserve these distinctions when interpreting the results.

4.1 Solar forecasts: completed single-cycle study

An 8,192-candidate search fitted 2025 data, selected settings on January–April 2026 and evaluated the frozen winners on May–August. Training used historical irradiance; later evaluation used the preceding day's forecast irradiance. The selected models were bidirectional GRU, Transformer, bidirectional LSTM and TCN respectively. Daylight errors are normalized by AC capacity:

Plant	Validation RMSE (%)	Later RMSE (%)	Later MAE (%)
1	4.708	6.069	4.006
345	7.645	10.366	6.893
901	10.241	11.231	6.776
910	8.977	10.992	6.630

These are retrospective errors against reviewed targets. For plant 910, 2,340 scored later quarters are synthetic; measured-only RMSE is 12.045%. That distinction is necessary to avoid interpreting reconstruction agreement as independent forecasting accuracy. The new four-cycle, 48-hour neural study is incomplete and has no final result yet.

Probabilistic results are mixed. Fixed calibrated quantiles beat the warm online combination at the day-ahead origin on all four plants in a 12-day August comparison. Later intraday updates help some plants. A separate January-fitted calibration worsens raw-ensemble CRPS on all four plants during a two-day August transfer. Calibration and online adaptation therefore need seasonal and decision-specific evaluation.

4.2 Price forecasts: useful baselines and modest incremental gains

In a 90-day January–March 2026 comparison, day-ahead elastic net with fundamentals and gas achieves MAE 8.11 EUR/MWh, versus 16.57 for previous-week persistence. For intraday 1 after day-ahead publication, however, simply using the day-ahead price gives MAE 4.90, better than the compared fitted spread model at 5.02 and direct model at 5.93. A strong simple baseline is essential.

In a separate native quarter-hour study, models chosen on June–August 2025 were followed with weekly refits on later complete September–November days:

Auction	Later MAE (EUR/MWh)	WAPE	Relative MAE
Intraday 1	6.88	6.10%	0.498
Intraday 2	7.12	6.26%	0.472
Intraday 3	10.54	9.08%	0.534

Relative MAE compares the same products with seasonal-lag persistence. The respective samples contain 8,160, 8,064 and 4,176 products. In a later 107-day day-ahead comparison, adding earlier-day intraday prices improves the earlier-selected model from 16.314 to 16.021 EUR/MWh. These studies retain assumed publication clocks and previously inspected periods; they do not yet establish live forecasting performance or a corresponding trading gain.

4.3 Trading and control: profit and imbalance can move together

Full January and August auction replays show that strict ensemble delivery can reduce absolute imbalance while reducing profit and increasing curtailment. For plant 1 in January, updating deterministic planning earns EUR 10,319.57 with 0.665 MWh absolute imbalance; strict 51-path planning earns EUR 10,054.81 with 0.226 MWh. Ending SOC is 44.14% versus 50.17%, respectively. Actual auction prices are known in these planning experiments, and inventory is not valued in cash profit.

A newer two-day January 15–16 control study completed 26 cases across four plants. The table reports *mean daily portfolio differences*, summing all four plants and comparing each case with its own tracking baseline:

Controller	Profit EUR/day	Abs. imbalance MWh/day	Wear EUR/day
Deterministic penalty MPC	+61.83	+2.85	+8.91
Ensemble penalty MPC	+37.40	+4.56	+93.05
Adaptive raw ensemble MPC	+150.86	+11.78	+128.26

The adaptive economic objective differs from the penalty controllers; these rows are illustrative policy outcomes, not a pure recourse ablation. The study has only two calendar days, common market prices across plants and reviewed production. Tracking sees actual interval PV feedback, whereas these MPC cases use forecast current PV. Later auction orders change with controller state. These differences prevent a general superiority claim.

Over an additional October 1–7 deterministic comparison, cash profit increases for all four plants, but total absolute imbalance increases 89.91 MWh and ending stored energy differs. On August 2–3 for plant 910, ensemble penalty control gains EUR 96.81 while the settlement-price oracle loses EUR 62.94. That August target is entirely reconstructed daylight or policy night zeros. Even an oracle with future prices is not an upper bound on realized profit when future solar production, terminal preferences and subsequent decisions differ.

The physical and cash-accounting checks passed for the completed reviewed January matrix, October week and August comparison. These checks establish internal consistency; they do not remove the empirical limitations above.

5 Current focus and unresolved questions

The central question is which forecast information is valuable *for the decisions that remain possible.* Our current work concentrates on four areas.

First, the new solar model study uses all four weather cycles over 48-hour horizons, with lead-time eligibility and a clear separation between historical irradiance used for learning and forecast irradiance available in operation. Second, price experiments examine earlier-cleared auction curves, recent training windows and robust losses using matched, selection-fixed follow-up. Some robust fits have convergence limitations, so completed fitting alone is not treated as a valid model ranking.

Third, adaptive scenario control tests how temporal uncertainty interacts with battery energy, power and connection limits. Fourth, continuation value and monetary risk examine decisions near horizon boundaries and in unfavorable scenarios. Computational reduction is evaluated within these matched problems.

The available evidence does not yet provide a full-year controller comparison, a joint calibrated law for solar production and every market price, or a selection-fixed economic comparison using the final new neural forecasts. Longer studies must preserve target provenance, comparable information sets, ending inventory and calendar-block uncertainty. Annual operating value, investment returns and deployment claims remain beyond the completed evidence.

A Full residual auction optimization

A.1 Sets, parameters and signs

Let $t \in \mathcal{T}$ index the remaining physical intervals of a delivery day, Δ_t their duration in hours, $s \in \mathcal{S}$ scenarios with probabilities w_s, and $a \in \mathcal{A}_k$ auctions still open at decision k. $\mathcal{T}_a$ is auction a's eligible delivery set. Native products group physical intervals; their identifiers retain daylight-saving offsets.

Solar availability is $V_{s,t}$ MW AC, battery capacity E MWh, power limit $\bar{P}$ MW, energy bounds $E^{\min}, E^{\max}$, efficiencies η_c, η_d and hourly self-discharge σ. The grid limits are $P^{\text{imp}}, P^{\text{exp}}$. Q_t^{fix} is the signed MWh position from all earlier submitted quantities, including still-unpriced orders. Positive trade/grid signs mean sale/export. Auction price coefficients are $\pi_{a,s,t}$ in EUR/MWh.

Controls $c_{n,t}, d_{n,t}$ are charging/discharging MW; $z_{n,t}$ is a charging binary. Curtailment is $u_{m,t}$ MW, injection $i_{s,t}$ MW, grid exchange $g_{s,t}$ MW and internal energy $e_{s,t}$ MWh. $q_{a,n,t}$ is a signed auction adjustment in MWh. Node maps $n_t(s), n_a(s), m_t(s)$ encode observable history. Leaf values below refer to their node-owned variables, so indistinguishable histories cannot take different actions.

A.2 Complete base problem

For throughput wear $D_s = c_{\text{thr}} \sum_t (c_{s,t} + d_{s,t})\Delta_t$ and optional import cost $I_s = f \sum_t [-g_{s,t}]_+ \Delta_t$, the residual model is

$$\max \quad \sum_s w_s \left[\sum_{a \in \mathcal{A}_k} \sum_{t \in \mathcal{T}_a} \pi_{a,s,t} q_{a,n_a(s),t} - D_s - I_s - C_s^{\text{trade}} \right] \tag{1}$$

$$\text{s.t.} \quad i_{s,t} + u_{m_t(s),t} = V_{s,t}, \quad i_{s,t} \geq 0, \quad u_{m_t(s),t} \geq 0, \qquad \forall s,t, \tag{2}$$

$$g_{s,t} = i_{s,t} + d_{n_t(s),t} - c_{n_t(s),t}, \qquad \forall s,t, \tag{3}$$

$$-P^{\text{imp}} \leq g_{s,t} \leq P^{\text{exp}}, \qquad \forall s,t, \tag{4}$$

$$0 \leq c_{n,t} \leq \bar{P} z_{n,t}, \quad 0 \leq d_{n,t} \leq \bar{P}(1 - z_{n,t}), \quad z_{n,t} \in \{0,1\}, \qquad \forall n,t, \tag{5}$$

$$e_{s,t+1} = (1-\sigma)^{\Delta_t} e_{s,t} + \eta_c c_{n_t(s),t} \Delta_t - d_{n_t(s),t} \Delta_t / \eta_d, \qquad \forall s,t, \tag{6}$$

$$E^{\min} \leq e_{s,t} \leq E^{\max}, \quad e_{s,0} = e^k, \qquad \forall s,t, \tag{7}$$

$$Q_{s,t} = Q_t^{\text{fix}} + \sum_{a : t \in \mathcal{T}_a} q_{a,n_a(s),t}, \qquad \forall s,t, \tag{8}$$

$$g_{s,t} \Delta_t = Q_{s,t}, \qquad \forall s,t, \tag{9}$$

$$-P^{\text{imp}} \Delta_t \leq q_{a,n,t} \leq P^{\text{exp}} \Delta_t, \qquad \forall a,n,t \in \mathcal{T}_a, \tag{10}$$

$$-P^{\text{imp}} \Delta_t \leq Q_t^{\text{fix}} + \sum_{b \leq a : t \in \mathcal{T}_b} q_{b,n_b(s),t} \leq P^{\text{exp}} \Delta_t, \qquad \forall a,s,t, \tag{11}$$

$$q_{a,n,t}/\Delta_t = q_{a,n,t'}/\Delta_{t'}, \qquad t,t' \text{ in the same native product.} \tag{12}$$

Only eligible quantities exist. The last equality ties average power within a native product; on equal quarter-hours it ties MWh shares directly. Fixed initial energy can be observed or a forecast projection to the eligible boundary. A configured terminal equality $e_{s,T} = e_F$ and/or floor $e_{s,T} \geq e_{\text{floor}}$ is added where the study requires it. A planning terminal target does not itself force an algebraic feedback controller to end at that realized energy.

For progressive transaction tariff F_m, previously submitted monthly gross volume V_m and new gross volume $G_{m,s} = \sum_{a : m(a) = m, t} |q_{a,n_a(s),t}|$,

$$C_s^{\text{trade}} = \sum_m \{F_m(V_m + G_{m,s}) - F_m(V_m)\}. \tag{13}$$

Exact absolute-value and piecewise-linear graphs are required if reachable marginal tariff bands are decreasing; a convex epigraph alone is insufficient. The month is the submission month, not

necessarily the delivery month. Earlier auction cash and incurred wear are sunk and absent from (1). Only the closing auction's order is submitted; other quantities are revisable plans. Sequential planning replaces $\mathcal{A}_k$ by the single current auction.

A.3 Stochastic deviation and exact-envelope variants

The stochastic alternative replaces exact delivery by

$$\delta^+_{s,t} - \delta^-_{s,t} = g_{s,t}\Delta_t - Q_{s,t}, \tag{14}$$

$$0 \le \delta^+_{s,t} \le M_t y_{s,t}, \quad 0 \le \delta^-_{s,t} \le M_t(1 - y_{s,t}), \quad y_{s,t} \in \{0,1\}, \tag{15}$$

where $M_t = (P^{\text{imp}} + P^{\text{exp}})\Delta_t$ or a tighter configured cap. It adds signed forecast settlement less $\kappa \sum_t (\delta^+ + \delta^-)$ to each path's profit. With $A_s = \sum_t (\delta^+ + \delta^-)$, optional tail-volume cost is $\rho[z + (1-\alpha)^{-1} \sum_s w_s \xi_s]$, with $\xi_s \ge A_s - z$, $\xi_s \ge 0$, $z \ge 0$. Zero planning settlement and positive volume deterrence distinguish risk planning from realized-price exploitation.

Under exact delivery, common trades/battery controls, identical price paths and compatible instantaneous curtailment recourse, i_t is shared and

$$0 \le i_t \le \min_s V_{s,t}, \qquad u_{s,t} = V_{s,t} - i_t. \tag{16}$$

This gives an exact reduced feasible set and objective. It does not apply to general adaptive or imbalance-enabled problems, and discarding low-probability paths is not part of the equivalence.

B Battery wear, control and realized accounting

B.1 Wear models

For replacement value R, reference internal cycle depth d_0, life N and capacity E, AC throughput wear can be calibrated as

$$c_{\text{thr}} = \frac{R}{N d_0 E(1/\eta_c + \eta_d)}, \qquad D = c_{\text{thr}} \sum_t (c_t + d_t)\Delta_t. \tag{17}$$

An optional segmented depth model uses stress $\Phi(d) = \alpha d^\beta$, $\beta \ge 1$, J segments and marginal discharge cost

$$m_j = \frac{RJ}{\eta_d E}\{\Phi(j/J) - \Phi((j-1)/J)\}, \qquad D = \sum_{t,j} m_j d_{j,t} \Delta_t. \tag{18}$$

Segment energy transitions sum to aggregate energy; shallow-first charging and discharging priorities preserve cycle-depth memory. The segment vector carries between decisions and cannot generally be recovered from SOC alone. This is an approximation to cycle wear, not dynamic capacity fade or thermal/calendar aging [1].

B.2 Physical MPC

For already submitted position Q^{sub}_t, utilization $a_{n,t} \in [0,1]$ and forecast solar $V_{s,t}$,

$$g_{s,t} = a_{n(s,t),t} V_{s,t} + d_{n(s,t),t} - c_{n(s,t),t}, \qquad b_{s,t} = g_{s,t}\Delta_t - Q^{\text{sub}}_t. \tag{19}$$

All physical and information-sharing constraints in Appendix A apply, but trade variables are absent. With forecast settlement $R_s = \sum_t (\hat{\lambda}^+_{s,t} [b_{s,t}]_+ - \hat{\lambda}^-_{s,t} [-b_{s,t}]_+)$, the economic problem is

$$\min \quad \sum_s w_s \left[D_s + I_s + K_s - R_s + \kappa \sum_t |b_{s,t}| \right] + \lambda \left[\eta + \frac{1}{1-\alpha} \sum_s w_s \xi_s \right], \tag{20}$$

$$\xi_s \geq D_s + I_s - R_s - \eta, \qquad \xi_s \geq 0, \quad \eta \in \mathbb{R}. \tag{21}$$

Price-blind control omits R_s and monetary risk, using a deviation penalty. Ordinary ensemble control shares every horizon control; adaptive control shares commands only while observed histories coincide. All paths share the first command. Only that command executes, followed by physical protection. The current physical state and eligible forecasts initialize the next solve.

For the settlement-price oracle, stage one minimizes expected absolute deviation. With feasible value U and lower bound L, stage two maximizes settlement less wear/import/terminal costs under $A \leq U(1+r) + b$. Certified stage-one solutions or explicitly gap-qualified incumbents are distinguished. Failed economic refinement retains the validated first-stage action. Actual future prices make this an information benchmark, not an operational forecast policy.

B.3 Feedback, chronology and profit

Commitment tracking uses target $g^*_t = Q^{\text{sub}}_t / \Delta_t$ and interval PV. Feasible charge and discharge are bounded by

$$C_t = \min\left(\bar{P}, \frac{[E^{\max} - \rho_t e_t]_+}{\eta_c \Delta_t} \right), \quad D_t = \min\left(\bar{P}, \frac{\eta_d \left[\rho_t e_t - E^{\min}\right]_+}{\Delta_t} \right). \tag{22}$$

It adjusts net battery power toward the grid target, recovers curtailed PV where headroom permits, and enforces connection limits. MPC instead issues its optimized first action, subject to physical protection. Interval-average PV feedback is an idealized information assumption; it differs from a causal forecast-only current-step MPC input.

Realized accounting uses

$$\delta_t = g^{\text{real}}_t \Delta_t - Q^{\text{sub}}_t, \tag{23}$$

$$J^{\text{real}} = \sum_{a,t} \pi^{\text{real}}_{a,t} q^{\text{sub}}_{a,t} + \sum_t \left(\lambda^+_t [\delta_t]_+ - \lambda^-_t [-\delta_t]_+ \right) - D^{\text{real}} - I^{\text{real}} - C^{\text{trade,real}}. \tag{24}$$

With a single settlement price this reduces to $\lambda_t \delta_t$. Forecast penalties, tail-risk terms and terminal continuation values are absent from realized cash. Final energy is reported separately. The state and binding positions carry across days; no daily physical reset occurs in a continuous comparison.

C Forecast learning, dependence and controlled comparisons

C.1 Learning and scoring

Normalized solar prediction fits weighted squared/Huber losses to reviewed targets, with preprocessing fitted only on training data and mandatory night zeros. Price regression fits standardized lag/fundamental predictors with squared or robust loss and optional L^1/L^2 coefficient penalties. Quantile regression minimizes $\rho_\tau(r) = r(\tau - \mathbf{1}\{r < 0\})$; distributional networks minimize likelihood or a proper distribution score. For ensemble values x_m, weights w_m and observation y,

$$\text{CRPS} = \sum_m w_m |x_m - y| - \frac{1}{2} \sum_{m,j} w_m w_j |x_m - x_j|. \tag{25}$$

This measures marginal predictive quality, not economic value or serial dependence. RMSE, MAE, coverage, bias and provenance-specific scores remain separate diagnostics. Price WAPE divides pooled absolute error by pooled absolute observed price, while relative MAE compares the same products with a named naive forecast.

C.2 Dependence and reduced scenarios

For raw member rank $r_{s,t}$ and ordered calibrated samples $z_{(j),t}$, ensemble copula coupling sets $V_{s,t}^{\mathrm{ECC}} = z_{(r_{s,t}),t}$ [2]. Independent leadwise permutations preserve marginals while removing serial structure; paired PV/price samples must be permuted together. Adaptive nodes use observed prefixes through a stated lag, never the identity of a future realization. A restricted observation tree is not perfect foresight.

Scenario reduction chooses representative paths using

$$\min_{|\mathcal{M}|=K} \sum_{s} w_s \min_{j\in\mathcal{M}} \|\phi_s - \phi_j\|_2, \qquad \tilde{w}_j = \sum_{s:\,\mathrm{assign}(s)=j} w_s. \tag{26}$$

Features ϕ_s summarize forecast power, ramps, cumulative balance/deficit, physical saturation and monetary exposure. Greedy selection approximates this criterion and has no general guarantee on trading-value loss.

C.3 Continuation and interaction effects

A nominal forecast dynamic program on stored energy estimates

$$W_t(e) = \max_{a\in\mathcal{A}(e)} \{r_t(e,a) + W_{t+1}(e')\}. \tag{27}$$

Its interpolated terminal value enters physical control as $-W_T(e_T)$. This approximation omits unknown future auction opportunities and full aging memory; it must not be interpreted as a realized payment.

For latest/frozen information I and all-auction/DA-only access A, the matched interaction is $J_{11} - J_{10} - J_{01} + J_{00}$. For an altered asset setting b, calibration–flexibility interaction is $(J_{\mathrm{cal},b} - J_{\mathrm{raw},b}) - (J_{\mathrm{cal},0} - J_{\mathrm{raw},0})$. These contrasts require common dates, targets, starting state and otherwise matched assumptions. Calendar blocks shared across plants are the appropriate resampling unit when prices and weather are common. The current two-day controller sample cannot support precise tail or significance claims.

References

[1] B. Xu et al., *Factoring the Cycle Aging Cost of Batteries Participating in Electricity Markets*, 2017. https://arxiv.org/abs/1707.04567.

[2] R. Schefzik, T. L. Thorarinsdottir and T. Gneiting, *Uncertainty Quantification in Complex Simulation Models Using Ensemble Copula Coupling*, Statistical Science 28(4), 616–640, 2013. https://arxiv.org/abs/1302.7149.